IMO 2026 — Autonomous Model Comparison

Independently graded results across four autonomous runs

Compiled July 2026 · all runs used one identical minimal agent harness · grading: independent verifier agents, IMO 0–7 scale, claims checked symbolically and by machine

Final graded scores

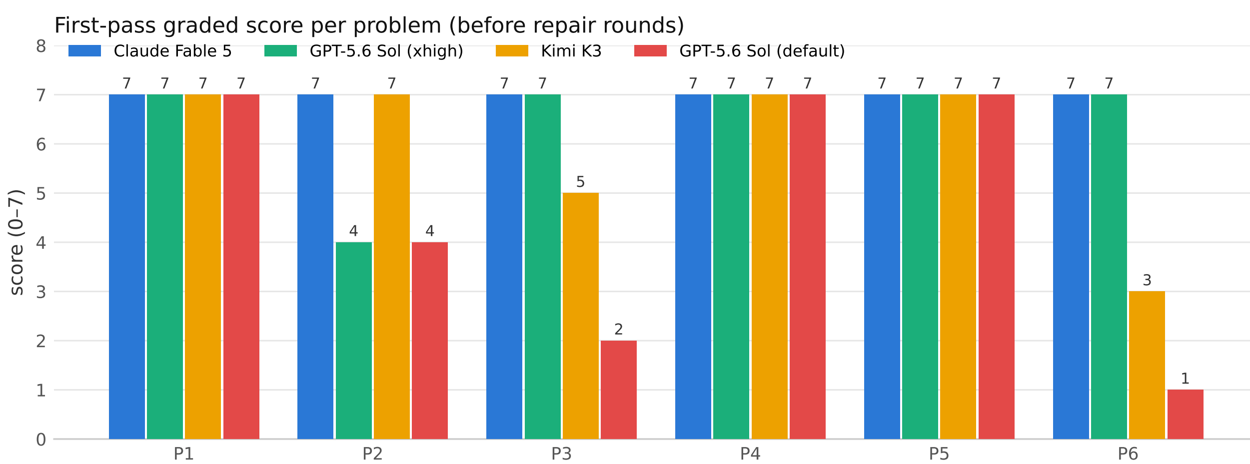
After repair rounds, in which a run was shown *only* its reviewer's specific defect findings (never another solution) and given further sessions. Every repaired score was re-graded as adversarially as the original.

Run	P1	P2	P3	P4	P5	P6	**Total**
Claude Fable 5 (default `high` effort)	7	7	7	7	7	7	**42/42**
GPT-5.6 Sol (`xhigh` effort)[†]	7	7	7	7	7	7	**42/42**
Kimi K3 (default effort)[†]	7	7	7	7	7	7	**42/42**
GPT-5.6 Sol (default effort)	7	4	2	7	7	1	**28/42**

[†] reached via repair rounds. Claude Fable 5 needed none — it is the only run graded 42/42 on its first pass.

First pass (no repairs)

Run	P1	P2	P3	P4	P5	P6	**Total**
Claude Fable 5	7	7	7	7	7	7	**42/42**
GPT-5.6 Sol (`xhigh`)	7	4	7	7	7	7	**39/42**
Kimi K3	7	7	5	7	7	3	**36/42**
GPT-5.6 Sol (default)	7	4	2	7	7	1	**28/42**



Time and cost (all rounds included)

Run	Final	Total time	Cost (USD)	Output tokens
Claude Fable 5	42/42	1.8 h	$38.83	371k
GPT-5.6 Sol (`xhigh`)	42/42	3.8 h	~$20.54	236k
Kimi K3	42/42	17.4 h	~$31.40	1,545k
GPT-5.6 Sol (default)	28/42	1.0 h	~$4.04	80k

Claude Fable 5 cost as metered by the Anthropic API. Kimi K3 estimated from logged tokens at list pricing ($0.30/M cache-hit, $3/M miss, $15/M output, 93% hit assumption); GPT-5.6 Sol from OpenRouter list pricing ($5/M in, $30/M out). Times include every retry and repair round.

Findings

- **Claude Fable 5 was perfect first-pass**, and fastest (1.8 h) — full rigor in a bare single-context loop with no reviewer, no orchestration, and no second chances, at its default effort. It is the only run that needed no repairs.
- **Reasoning effort buys rigor and calibration**: GPT-5.6 Sol went 28→39/42 on first pass moving from default to `xhigh` in the identical harness. Its P3 and P6 became verified 7s (a previously *false* lemma replaced by a correct argument), and its P2 failure turned into an honestly self-declared partial rather than a confident invalid proof.
- **Defect-only feedback closes the remaining gap**: given nothing but a reviewer's named defects, GPT-5.6 Sol (`xhigh`) repaired P2 and Kimi K3 repaired P3 and its P6 finiteness crux — inventing a new compactness theorem from the greedy killing property — bringing three of four runs to a verified 42/42.
- **Self-reports inflate**: nearly every run claimed every problem "solved"; grading confirmed far fewer, with each deduction backed by a named, checkable defect (exact minimax refutations, explicit counterexample families, symbolic identity checks).

Caveats

- Graders are Claude-based agents, not human medalists; treat scores as strong but not authoritative.
- All four runs used the identical harness (same system prompt, tools, time caps), so they are directly comparable. Reasoning settings: Claude Fable 5 at its API default effort (`high`, always-on adaptive thinking); Kimi K3 at its API default; GPT-5.6 Sol at default (`medium`) and `xhigh` — the one controlled difference between its two runs.
- Repair rounds are marked; the first-pass table above is the clean single-shot comparison.
- Full per-turn audit trails for every run and round, plus per-problem grader verdicts with named defects, accompany this document.